Comparing Air Pollution Trends Between Delhi, India and Dhaka, Bangladesh from 2003-2023

ENV 872L Final Project

Aadya Shukla and Humayra Rahman

2026-04-24

Contents

Introduction . . . 3
Data description . . . 3
- Dhaka Dataset . . . 3
- Delhi Dataset . . . 3

Methods . . . 4
- Initial Data Wrangling . . . 4
- Visualization . . . 4
- Time Series . . . 4
- Trend Analysis . . . 4
- Comparative Analysis . . . 4

Results . . . 5
Discussion . . . 11
- Hypothesis 1: Does Air Quality Have a Significant Seasonal Component and Decreasing Trend? 11
- Hypothesis 2: Is Delhi's Air Quality Significantly Worse than Dhaka's? . . . 11
- Hypothesis 3: Is Delhi's Air Quality Decreasing More Rapidly than Dhaka's? . . . 11

Limitations . . . 12
Conclusion . . . 12
References . . . 12

List of Figures

1 Monthly AQI Trend from 2003-2023 . . . 5
2 Monthly PM 2.5 Trend from 2003-2023 . . . 6
3 Monthly PM 10 Trend from 2003-2023 . . . 7

4 Monthly Ozone Trend from 2003-2023 8
5 Decomposition of Delhi's AQI Time Series (2003-2023) 9
6 Decomposition of Dhaka's AQI Time Series (2003-2023) 10

List of Tables

1 Summary of Mann-Kendall Results 5
2 Summary of two sample t-test Results 11
3 Summary of Linear Regression Results 11

Introduction

Air pollution is one of the most pressing environmental and public health challenges of the 21st century, particularly across South Asia, where rapid urbanization, industrial growth, vehicular emissions, and agricultural waste burning have made several cities among the most polluted in the world (Ditta et al., 2026). Delhi, the capital of India, and Dhaka, the capital of Bangladesh, are two such megacities that consistently rank among the worst for air quality globally, with both cities experiencing severe seasonal pollution cycles driven by winter temperature inversions, stubble burning, brick kilns, and vehicle emissions along the shared Indo-Gangetic Plain (Ditta et al., 2026; Begum et al., 2018). Despite their geographic proximity and shared pollution sources, the two cities have followed divergent regulatory paths. India launched its National Clean Air Programme in 2019 targeting significant reductions in particulate matter (Garg et al., 2022), while Bangladesh was among the first South Asian nations to ban two-stroke engine vehicles in the early 2000s and transition to CNG-powered alternatives (State of Global Air, n.d.), raising important questions about whether these differing policy trajectories have translated into meaningfully different pollution trends over time.

While prior studies have individually examined air quality trends in Delhi (Garg et al., 2022; Singh et al., 2023) and Dhaka (Islam et al., 2021; Begum & Hopke, 2018), long-term comparative analyses spanning both cities remain limited. This project addresses that gap by using 20 years of monthly air quality data (2003 to 2023) to compare air quality trends in PM2.5, PM10, ozone (O3), and AQI across both cities. This study addresses the following hypotheses:

H_1: Air quality in Delhi and Dhaka exhibits a statistically significant seasonal trend over the 2003–2023 study period.

H_2: Air quality in Delhi and Dhaka exhibits a statistically significant long-term decreasing trend over the 2003–2023 study period.

H_3: Delhi's air quality is significantly worse than Dhaka's, as measured by mean concentrations of PM2.5, PM10, O3, and AQI over the 2003–2023 study period.

Data description

Dhaka Dataset

The Dhaka air quality data was sourced from the Bangladesh Air Quality Index (AQI) Dataset (2000–2025), published by Tapon Paul of Daffodil International University and retrieved from Mendeley Data (Paul, 2026). The full dataset contains 1,048,551 hourly records spanning 103 cities across Bangladesh across 13 columns: city_id, city_name, lat, lon, datetime, pm10, pm2_5, carbon_monoxide, carbon_dioxide, nitrogen_dioxide, sulphur_dioxide, ozone, and aqi. AQI values are calculated according to the US EPA standard.

Delhi Dataset

The Delhi air quality data was sourced from the Processed Delhi Air Quality and Weather Data dataset curated by Abhinav Sarkar and retrieved from Hugging Face (Sarkar, 2024). The full dataset contains approximately 2.9 million hourly observations from multiple monitoring locations across Delhi from March 2000 to November 2024, across 15 columns: city, event_timestamp, temperature, humidity, pressure, wind_speed, wind_direction, pm25, pm10, no2, so2, o3, co, aqi, and location_id. The dataset was pre-processed prior to publication, with missing AQI values removed and remaining gaps addressed through mean imputation and nearest neighbor interpolation.

Methods

Initial Data Wrangling

Both datasets were cleaned and processed separately before being combined for analysis. The Dhaka dataset was filtered to retain only observations from Dhaka city. For both datasets, the datetime columns were converted into a proper R datetime object using the ymd_hms function and only the columns needed for analysis were selected: Location, Month, Year, PM2.5, PM10, O3, and AQI. Both datasets were filtered to January 1, 2003 through December 31, 2023. We chose this window because the Dhaka dataset spans a longer date range from 2000 to 2025 and the Delhi dataset does not have complete monthly records for 2000 and 2024. To keep the data consistent and comparable across both cities, we focused on a clean 20-year trend. Hourly observations for both datasets were aggregated into monthly averages for all four pollutants (AQI, PM 2.5, PM 10, and O3) using na.rm = TRUE to handle any missing values. A Location variable was added to each dataset before using rbind to combine the two datasets. A date column marking the first day of each month was created using make_date for use in visualizations and regression analyses. The final combined dataset contained 504 average monthly observations over 8 columns (see below).

```
## 'data.frame':    504 obs. of  8 variables:
##  $ month      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ year       : int  2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 ...
##  $ PM2.5      : num  70 74.6 82.5 88.9 95.5 ...
##  $ PM10       : num  119 127 139 151 162 ...
##  $ O3         : num  42.4 42.8 43.2 42.5 42.6 ...
##  $ AQI        : num  151 156 162 166 170 ...
##  $ Location   : chr  "Dhaka" "Dhaka" "Dhaka" "Dhaka" ...
##  $ first.month: Date, format: "2003-01-01" "2004-01-01" ...
```

Visualization

To get an initial sense of how pollution has changed over time in each city, we plotted monthly trends for all four pollutants. Both cities were displayed on the same graph with a linear trend line to highlight the overall direction of change across the 20-year study period.

Time Series

We built monthly time series objects for each pollutant and each city. Each series was then decomposed into the seasonal, long-term trend, and remainder components.

Trend Analysis

To test whether pollution levels were changing significantly over time, we ran both seasonal and non-seasonal Mann-Kendall tests for each pollutant in each city. We chose to run both because the initial graphs suggested a more likely seasonal trend in Dhaka's air quality with weaker seasonality in Delhi's air quality. The non-seasonal Mann-Kendall test was run on the de-seasonalized series, obtained by subtracting the seasonal component from the STL decomposition.

Comparative Analysis

Before running any comparative analyses, we had to check if our data was normally distributed. For this, we generated some qq plots for all four variables for each city. Then, to compare the air quality between the two cities, we used two approaches. First, we ran two-sample t-tests for each pollutant to test whether mean

pollution levels over the full study period were significantly different between Delhi and Dhaka. Second, we fitted linear regression models with an interaction term between time and location for each pollutant, using the formula lm(pollutant ~ first.month * Location).

Results

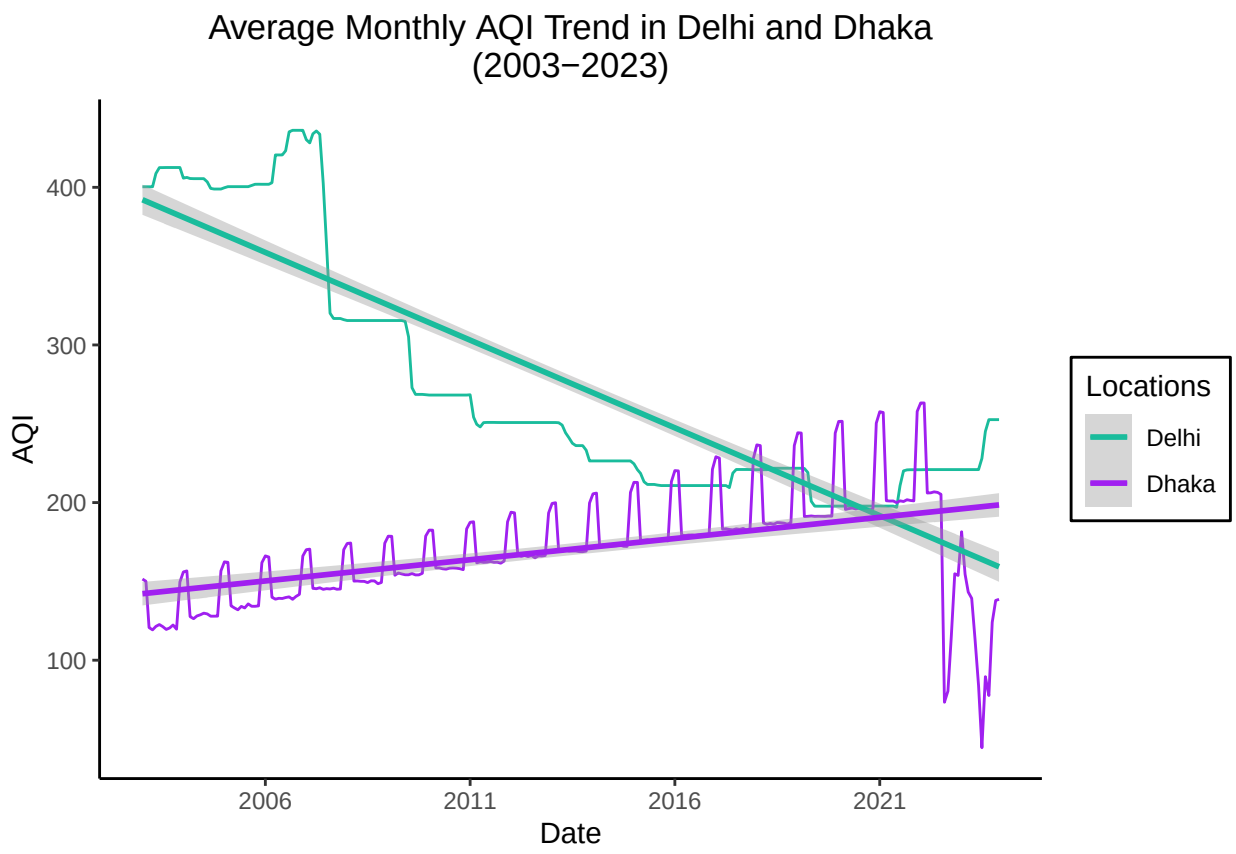

Figure 1: Monthly AQI Trend from 2003-2023

Table 1: Summary of Mann-Kendall Results

Location	Variable	Seasonal p-value	Non-Seasonal p-value	Trend Statistic
Dhaka	AQI	0.0768	0.2045	-204
Dhaka	PM 2.5	0.0768	0.2354	-204
Dhaka	PM 10	0.1068	0.2354	-186
Dhaka	O3	**1.43e-06**	**2.06e-06**	-554
Delhi	AQI	0.3226	0.1078	-114
Delhi	PM 2.5	0.0862	**0.0330**	-197
Delhi	PM 10	0.4257	0.1409	-92
Delhi	O3	0.3098	0.4421	117

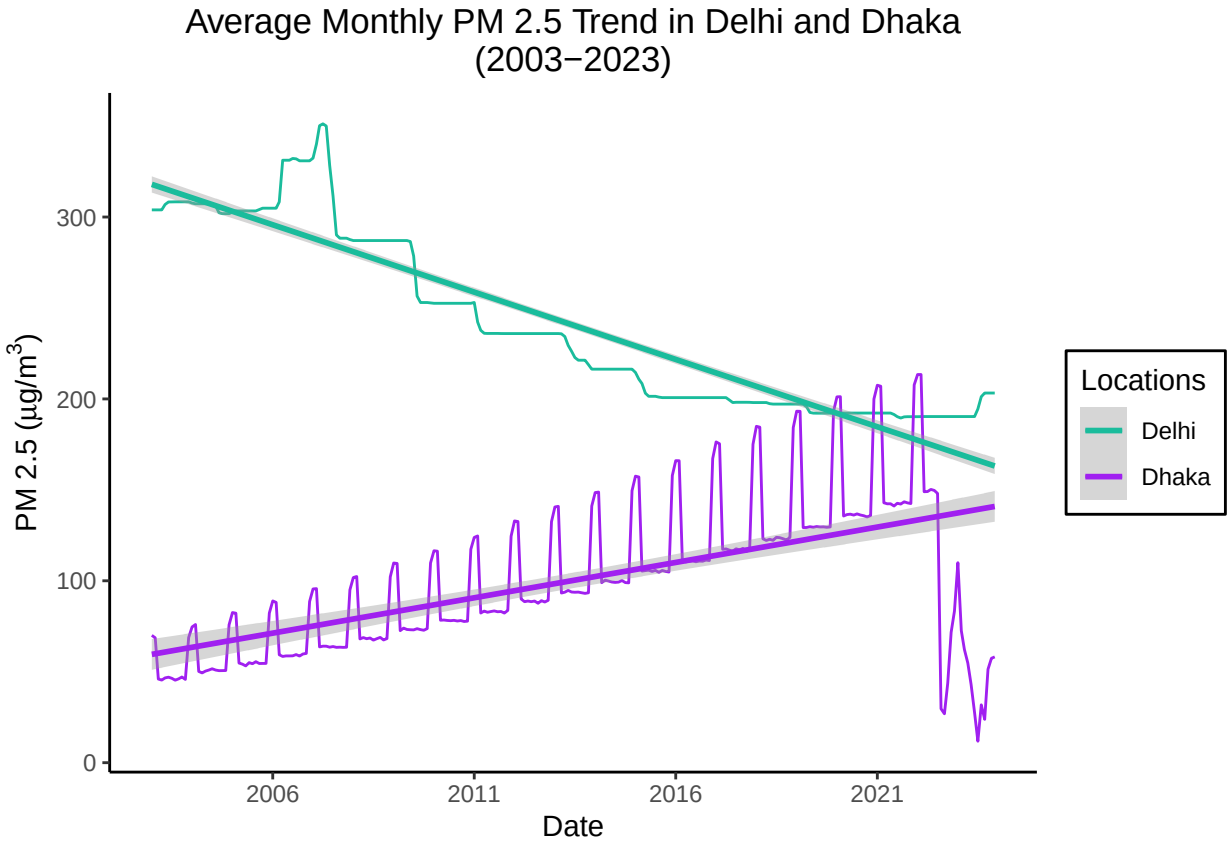

Figure 2: Monthly PM 2.5 Trend from 2003-2023

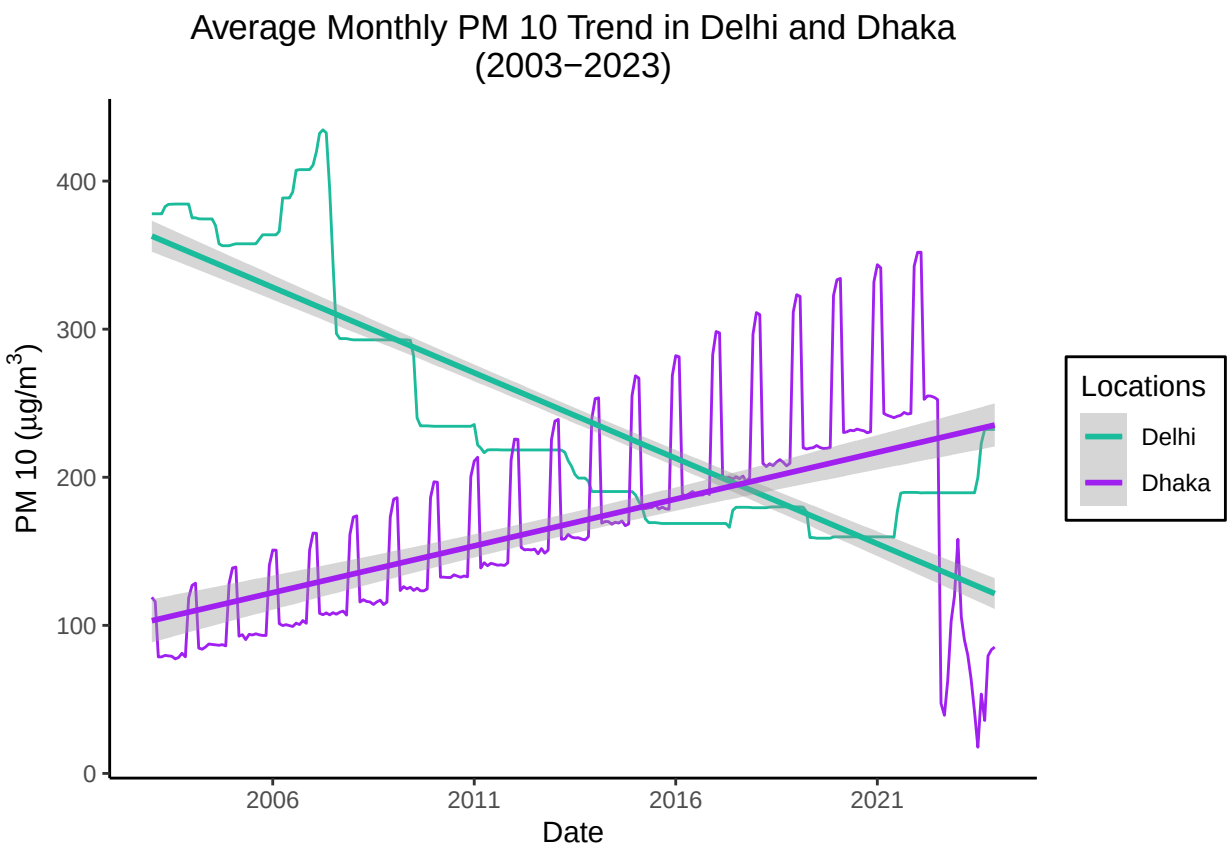

Figure 3: Monthly PM 10 Trend from 2003-2023

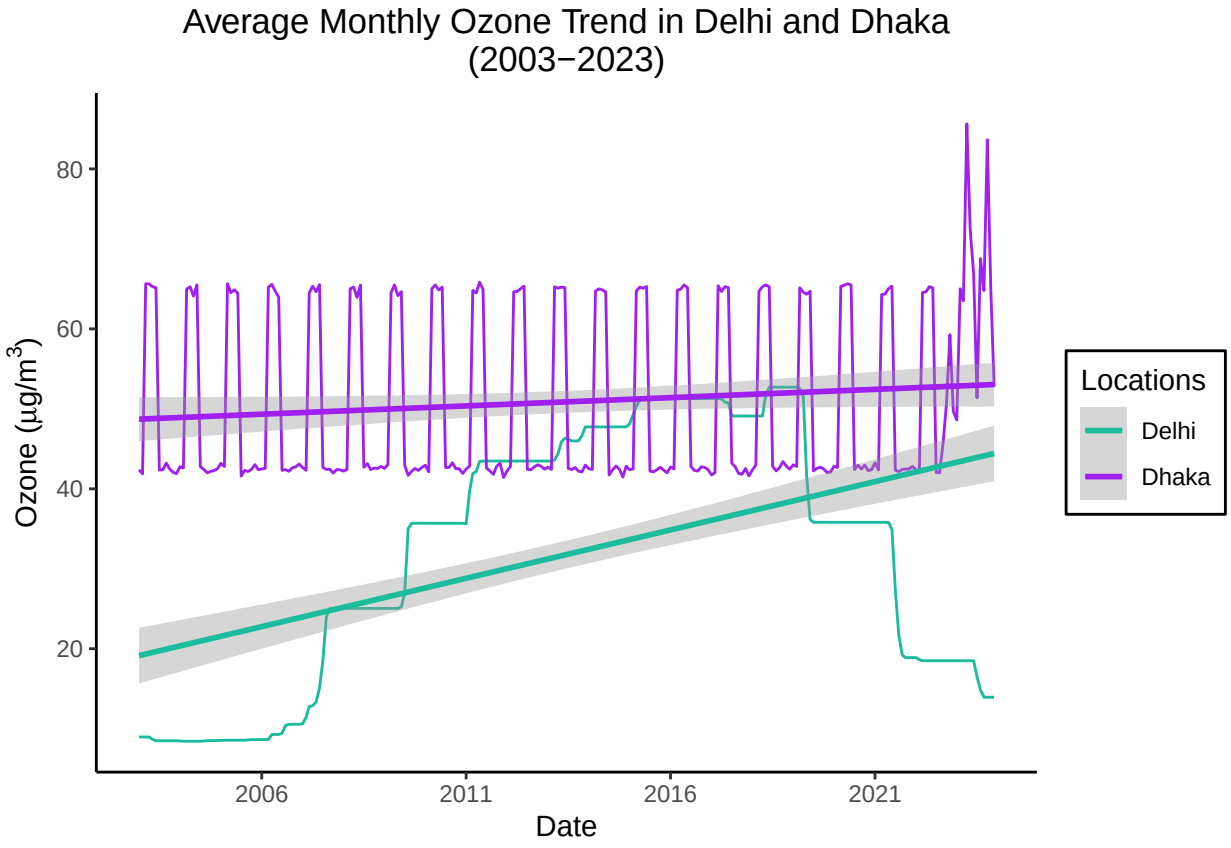

Figure 4: Monthly Ozone Trend from 2003-2023

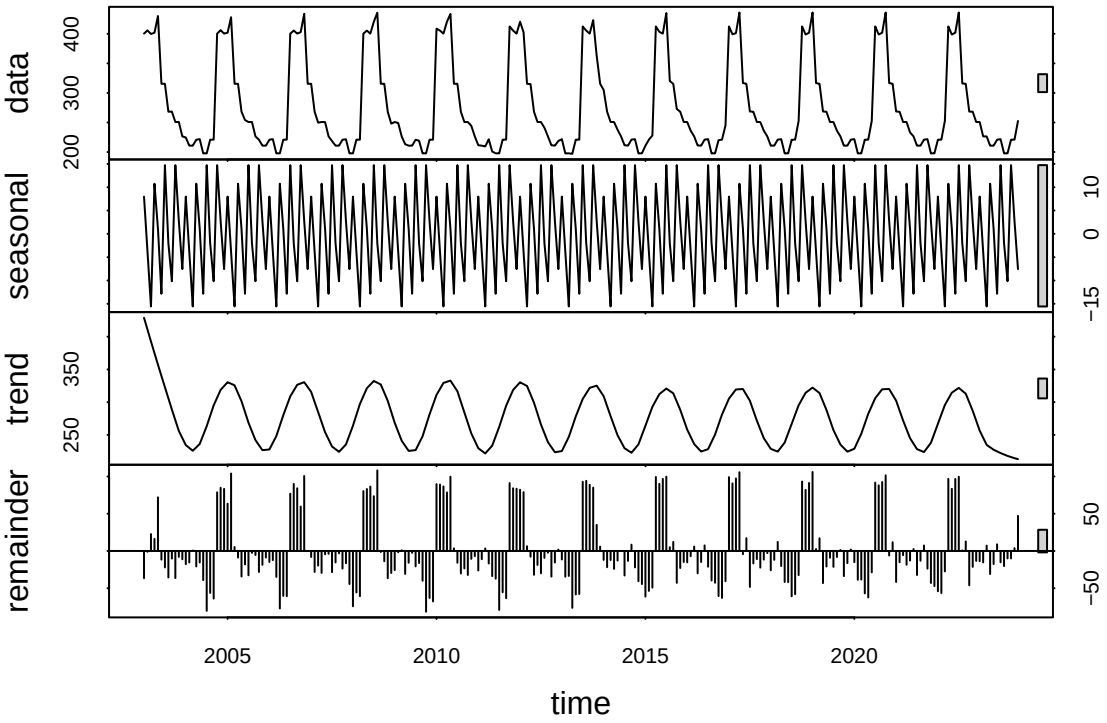

Figure 5: Decomposition of Delhi's AQI Time Series (2003-2023)

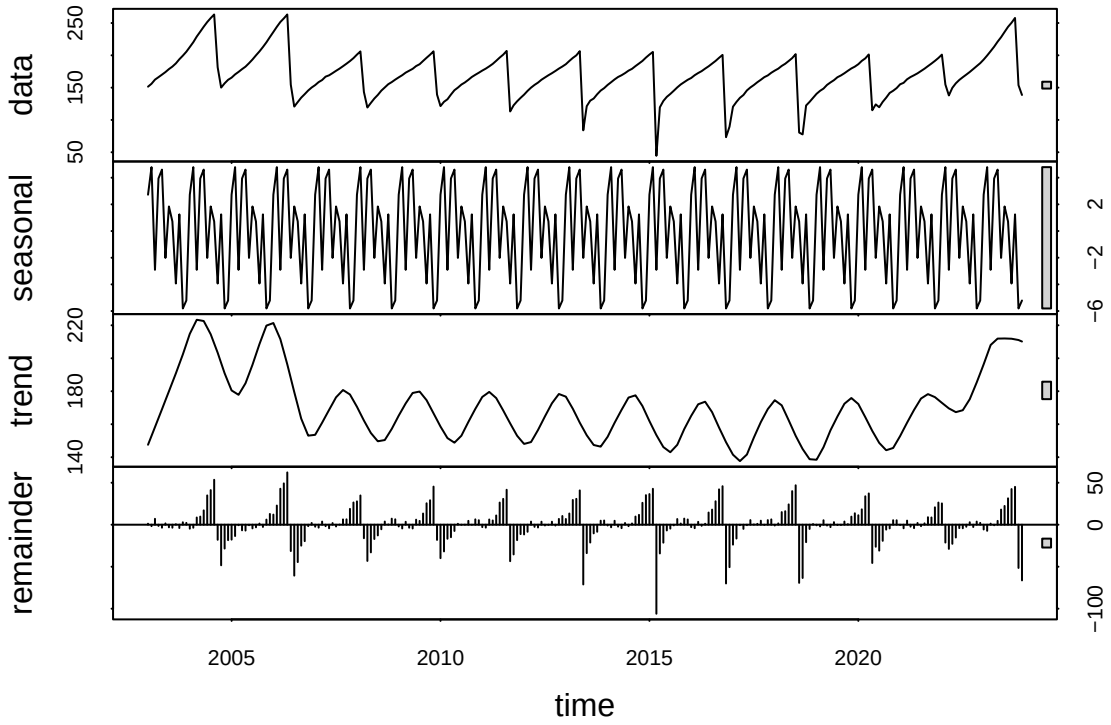

Figure 6: Decomposition of Dhaka's AQI Time Series (2003-2023)

Table 2: Summary of two sample t-test Results

Variable	p-value	Dhaka mean	Delhi mean
AQI	**2.2e-16**	170.35	275.71
PM2.5	**2.2e-16**	100.21	240.54
PM10	**2.2e-16**	169.18	242.17
O3	**2.2e-16**	50.87	31.78

Table 3: Summary of Linear Regression Results

Variable	p-value	Min	Median	Max	R-squared
AQI	**2.2e-16**	-152.894	-1.296	93.331	0.812
PM2.5	**2.2e-16**	-127.439	-3.511	79.986	0.894
PM10	**2.2e-16**	-214.902	-5.708	128.747	0.634
O3	**2.2e-16**	-30.487	-4.640	32.740	0.424

Discussion

Hypothesis 1: Does Air Quality Have a Significant Seasonal Component and Decreasing Trend?

The STL decomposition answered the seasonal part clearly. Both cities show strong, predictable annual cycles, with pollution peaking every winter and dropping during the monsoon without fail. This is expected given the Indo-Gangetic Plain's meteorology, where cold dry winters trap pollutants near the surface and monsoon rains wash them out (Begum et al., 2018; Ditta et al., 2026).

Finding a statistically significant decreasing trend was harder. The Mann-Kendall tests only confirmed significant declines in two out of eight combinations. Dhaka O3 was the clearest result, with both tests agreeing on a highly significant decline. Delhi PM2.5 only crossed the significance threshold after removing the seasonal component, suggesting the decline is real but gets partially buried in seasonal noise. Everything else came back non-significant. This does not mean pollution is not changing as Delhi's particulate decline is visually obvious, but it does mean the changes have not been consistent enough across all months and years to be statistically confirmed. That distinction between a visible trend and a statistically significant one is important when interpreting these results.

Hypothesis 2: Is Delhi's Air Quality Significantly Worse than Dhaka's?

For particulate pollution, yes. Delhi's mean AQI, PM2.5, and PM10 were all significantly higher than Dhaka's across the full study period ($p < 2.2e\text{-}16$ for all three), which is consistent with Delhi's reputation as one of the most polluted cities in South Asia (Singh et al., 2023). Ozone flipped this result entirely. Dhaka had significantly higher mean O3 than Delhi, which goes against the expectation that Delhi would simply be worse across the board. Unlike particulate matter, ozone forms through photochemical reactions rather than direct emissions, so the two cities having different ozone profiles is not necessarily contradictory. Dhaka's consistently higher ozone, combined with the unusual jump in values around 2007 to 2008, suggests either a change in monitoring methodology or a genuine shift in photochemical activity following fuel regulation changes in Bangladesh. Either way, it is a finding worth investigating further and it shows that framing Delhi as simply more polluted than Dhaka is an oversimplification.

Hypothesis 3: Is Delhi's Air Quality Decreasing More Rapidly than Dhaka's?

The linear regression gave the clearest answer here. The interaction term between time and location was significant in all four models, confirming that the two cities are changing at statistically different rates for

every pollutant. For AQI, PM2.5, and PM10, Delhi was declining while Dhaka was increasing. The models fit this pattern well, particularly for PM2.5 (R squared = 0.89) and AQI (R squared = 0.81), giving us confidence that time and location together genuinely explain most of the variation in particulate pollution. PM10 fit moderately (R squared = 0.63) and O3 fit less well (R squared = 0.43), which makes sense given how much more complex ozone formation is compared to direct particulate emissions. For O3, the roles reversed. Delhi's ozone was rising while Dhaka's was declining, and both the regression interaction term and the Mann-Kendall tests told the same consistent story. So to directly answer Hypothesis 3: yes, Delhi's particulate pollution is decreasing more rapidly than Dhaka's, and the gap has been narrowing throughout the study period. The PM10 lines have already crossed. All three analyses point to the same conclusion. Delhi and Dhaka are converging from opposite directions for particulate pollution. Delhi started far worse but its policy interventions, including stricter vehicle emissions standards, metro rail expansion, and the National Clean Air Programme, appear to have made a real difference (Garg et al., 2022). Dhaka, despite early gains from banning two-stroke engines, has seen particulate pollution worsen steadily as urbanization, vehicle growth, and brick kiln proliferation have outpaced regulatory capacity (State of Global Air, n.d.). If these trends continue, Dhaka is on track to approach Delhi level particulate pollution within the coming decades, and Bangladesh's National Air Quality Management Plan 2024 to 2030 will need to deliver meaningful results to change that trajectory (World Bank, 2022). Dhaka is the significant decline in ozone confirmed by both the Mann-Kendall tests and the regression model. However, given the unusual data patterns in both cities' O3 series, those results deserve some caution and would benefit from validation against independent monitoring sources.

Limitations

Conclusion

References